

IRIS Staging of CKD (Modified 2026)

1. Staging of Chronic Kidney Disease based on blood creatinine and SDMA concentrations

Staging is undertaken following diagnosis of chronic kidney disease (CKD) to facilitate appropriate treatment and monitoring of the canine or feline patient. Staging is based initially on fasting blood creatinine or fasting blood SDMA concentration or (preferably) both assessed on at least two occasions in a hydrated, stable patient. The dog or cat is then substaged based on proteinuria and blood pressure.

Using these criteria, some empirical recommendations can be made about the type of treatment it would be logical to use for these cases. In addition, predictions based on clinical experience might be made about the likely response to treatment.

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Stage	1		2		3		4	
	Dogs	Cats	Dogs	Cats	Dogs	Cats	Dogs	Cats
Blood creatinine* µmol/l	<125	<140	125 – 250	140 – 250	251 – 440	251 – 440	>440	>440
Blood creatinine* mg/dl	<1.4	<1.6	1.4 – 2.8	1.6 – 2.8	2.9 – 5.0	2.9 – 5.0	>5.0	>5.0
SDMA µg/dl	<18	<18	18 – 35	18 – 25	36 – 54	26 – 38	>54	>38

Comments

Stage 1

Normal blood creatinine or normal or mild increase blood SDMA. Some other renal abnormality present (such as, inadequate urinary concentrating ability without identifiable non-renal cause (in cats not dogs), abnormal renal palpation or renal imaging findings, proteinuria of renal origin, abnormal renal biopsy results, increasing blood creatinine or SDMA concentrations in samples collected serially). Persistently elevated blood SDMA concentration (>14 µg/dl) may be used to diagnose early CKD.

Stage 2

Normal or mildly increased creatinine, mild renal azotemia (lower end of the range lies within reference ranges for creatinine for many laboratories, but the insensitivity of creatinine concentration as a screening test means that patients with creatinine values close to the upper reference limit often have excretory failure). Mildly increased SDMA. Clinical signs usually mild or absent.

Stage 3

Moderate renal azotemia. Many extrarenal signs may be present, but their extent and severity may vary. If signs are absent, the case could be considered as early Stage 3, while presence of many or marked systemic signs might justify classification as late Stage 3.

Stage 4

Increasing risk of systemic clinical signs and uremic crises.

* The blood creatinine concentrations apply to average size dogs – those of extreme size may vary.

The recommendations for SDMA are based on published literature which utilizes proprietary IDEXX technology for measuring SDMA.

At this time, it is not known if other assays will provide equivalent results.

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Discrepancies between creatinine and SDMA

IRIS CKD staging is based on fasting blood creatinine concentration and blood SDMA concentration. SDMA is potentially a more sensitive marker that is less impacted by loss of lean body mass.

DOGS:

Stage and Treat as IRIS CKD Stage 2

If serum or plasma SDMA is persistently **>18 µg/dl** in a dog whose creatinine is **< 125 µmol/l [1.4 mg/dl]** (IRIS CKD stage 1 based on creatinine).

Stage and Treat as IRIS CKD Stage 3

If serum or plasma SDMA is persistently **>35 µg/dl** in a dog whose creatinine is between **125 and 250 µmol/l [1.4 and 2.8 mg/dl]** (IRIS CKD stage 2 based on creatinine).

Stage and Treat as IRIS CKD Stage 4

If serum or plasma SDMA is persistently **>54 µg/dl** in a dog whose creatinine is between **251 and 440 µmol/l [2.9 and 5.0 mg/dl]** (IRIS CKD stage 3 based on creatinine).

CATS:

Stage and Treat as IRIS CKD Stage 2

If serum or plasma SDMA is persistently **>18 µg/dl** in a cat whose creatinine is **<140 µmol/l [1.6 mg/dl]** (IRIS CKD stage 1 based on creatinine).

Stage and Treat as IRIS CKD Stage 3

If serum or plasma SDMA is persistently **>25 µg/dl** in a cat whose creatinine is between **141 and 250 µmol/l [1.6 and 2.8 mg/dl]** (IRIS CKD stage 2 based on creatinine).

Stage and Treat as IRIS CKD Stage 4

If serum or plasma SDMA is persistently **>38 µg/dl** in a cat whose creatinine is between **251 and 440 µmol/l [2.9 and 5.0 mg/dl]** (IRIS CKD stage 3 based on creatinine).

SDMA assays are offered by a number of laboratories throughout the world. The methodology used has not yet been standardized and the recommendations made above are based on the proprietary methodology offered by IDEXX Laboratories.

These recommendations are based on current state of knowledge where SDMA appears to be a more sensitive indicator of early-stage CKD in the dog and cat.

Data are starting to emerge on conditions where SDMA may be elevated without a reduction in GFR, where serum creatinine values are unaffected. Lymphoma in dogs and cats is one such example. This illustrates the value of having two biomarkers which are surrogate markers for GFR which can be interpreted together.

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Breed and size associated effects on serum creatinine and SDMA

Healthy Birman cats and greyhound dogs have been found to have higher serum SDMA values than other breeds. Both these breeds also have higher serum creatinine values, taking some healthy individuals (up to 20% of Birman cats) outside laboratory reference intervals.

2a. Substaging by Proteinuria

The goal is to identify renal proteinuria having ruled out post-renal and pre-renal causes.

Standard urine dipsticks can be insensitive therefore practitioners should consider using a more reliable quantitative test such as the urine protein to creatinine ratio (UP/C) or a species-specific albuminuria assay.

The UP/C should be measured in all dogs and cats with CKD, provided there is no evidence of urinary tract inflammation or hemorrhage, and the routine measurement of plasma proteins has ruled out dysproteinemias. Ideally sub-staging should be done on the basis of at least two urine samples collected over a period of at least 2 weeks.

UP/C value		Substage
Dogs	Cats	
<0.2	<0.2	Non-proteinuric
0.2 to 0.5	0.2 to 0.4	Borderline proteinuric
>0.5	>0.4	Proteinuric

Canine and feline patients that are persistently borderline proteinuric should be re-evaluated within 2 months and re-classified as appropriate.

The significance of borderline proteinuria in predicting future renal health is not completely understood. IRIS's recommendation is to continue to monitor this level of proteinuria in dogs. Veterinarians might offer treatment for cats with persistent borderline proteinuria considering the association with progressive kidney disease (see treatment guidelines).

Response to any treatment given to reduce glomerular hypertension, filtration pressure, and proteinuria, should be monitored at intervals using UP/C.

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2b. Substaging by Blood pressure

Canine and feline patients should be acclimatized to the measurement conditions and multiple measurements taken. The final classification should rely upon multiple systolic blood pressure determinations, preferably done during repeated patient visits to the clinic on separate days, but acceptable if during the same visit with at least 2 hours separating determinations. Patients are substaged by systolic blood pressure according to the degree of risk of target organ damage, and whether there is evidence of target organ damage or complications.

For most dogs and all cats, the IRIS blood pressure substages are as follows:

Systolic Blood Pressure mmHg	Blood Pressure Substage	Risk of Future Target Organ Damage
<140	Normotensive	Minimal
140 – 159	Prehypertensive	Low
160 – 179	Hypertensive	Moderate
≥ 180	Severely hypertensive	High

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However, some breeds of dog, particularly sight hounds, tend to have higher blood pressure than other breeds. It is preferable to use breed-specific reference ranges if available. The classification of risk of future target organ damage in “high-pressure breeds” might be adjusted as follows:

Minimal risk	systolic pressure <10 mm Hg above the breed-specific reference interval
Low risk	systolic pressure 10-20 mm Hg above the breed-specific reference interval
Moderate risk	systolic pressure 20-40 mm Hg above the breed-specific reference interval
High risk	systolic pressure >40 mm Hg above the breed-specific reference interval.

As with proteinuria, in the absence of evidence of existing target organ damage, demonstration of persistence of blood pressure readings within a particular category is important. ‘Persistence’ of increased systolic blood pressure should be judged on multiple measurements made at different clinic visits (within 1-2 weeks). The higher the systolic blood pressure, the shorter the time frame over which persistence should be judged.

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3. Revision of staging and substaging after treatment

The stage and substages assigned to the patient should be revised appropriately as changes occur. For example, a substantial increase in blood creatinine or SDMA concentration might warrant reassignment to a higher stage to reflect the new situation.

Similarly, if antihypertensive (or antiproteinuric) treatment has been instituted, the patient's classification on re-evaluation should be adjusted if necessary to reflect the new blood pressure (or UP/C) rather than the original status, with the addition of an indication that the current classification is affected by treatment.

The following two examples illustrate the process of revision, where 'treating' is used as an indicator of ongoing treatment.

Example 1

Euvolemic cat with stable renal function

Creatinine 200 µmol/l [2.3 mg/dl]
SDMA 22 µg/dl
UP/C 0.32
Systolic blood pressure 200 mm Hg
Classification – *IRIS CKD Stage 2, borderline proteinuric, severely hypertensive.*

Same cat after antihypertensive treatment

Creatinine 220 µmol/l [2.5 mg/dl]
SDMA 24 µg/dl
UP/C 0.12
Systolic blood pressure 155 mm Hg
New classification – *IRIS CKD Stage 2, non-proteinuric, prehypertensive (treating).*

Example 2

Euvolemic dog with stable renal function

Creatinine 230 µmol/l [2.6 mg/dl]
SDMA 39 µg/dl
UP/C 0.8
Systolic blood pressure 155 mm Hg
Classification – *IRIS CKD Stage 3, proteinuric, prehypertensive*

*Note: As described above in the **Discrepancies between creatinine and SDMA** section, if blood SDMA is persistently >35 µg/dl in a canine patient whose blood creatinine is between 125 and 250 µmol/l (1.4 and 2.8 mg/dl; IRIS CKD stage 2 based on creatinine), this dog should be staged and treated as an IRIS CKD Stage 3 patient.*

Same dog after antiproteinuric treatment

Creatinine 240 µmol/l [2.7 mg/dl]
SDMA 42 µg/dl
UP/C 0.4
Systolic blood pressure 155 mm Hg
New classification – *IRIS CKD Stage 3, borderline proteinuric (treating), prehypertensive.*